

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements: *coherence*, meaning collective phase order; *compatibility*, meaning that local descriptions agree where their domains overlap; and *self-representation*, meaning that the resulting state carries an accurate reconstruction of itself. The first supplies neither of the others, and four results establish that without any cortical hypothesis. Synchrony constrains content only through encoders shared on overlaps, and then by the resultant of the patch the bound is read on rather than of the whole population. The resultant does not determine the state: a winding field and a uniform one are stationary at one coupling with resultants spanning nearly the range of that observable. A contraction with a unique fixed point can discard its input, so accuracy across a family of states requires distinguishable codes. And where the encoding is a region's restriction, that resolution is demanded of the region before any readout is chosen. Under stated hypotheses, compatible local states then glue to a unique global state that an accurate contracting reconstruction fixes uniquely, and installed energy bounds the mean coupling a declared actuator supplies, so a coherent stationary branch requires it

to exceed $2D/\kappa$, with D the phase diffusion and κ set by the mode prices; energy is not the scarce side of that condition for a macroscopic substrate, which leaves it a statement about κ , derived here for neither cortex nor a processor. Lean 4 checks these results and their composition, with the eight connecting hypotheses as explicit arguments and no axioms beyond its libraries'. Further controls bound what observation can decide: distinct mechanisms share stationary phase distributions, and a coupling rule descending a squared-drift objective holds coherence while losing the structure it was meant to acquire. One prediction needs no absolute calibration: onset delay scales as the square root of the rate at which coupling crosses threshold, measured here as $\Delta K \propto v^{0.444}$, which a within-subject emergence-from-anaesthesia design can test. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling. Testing it requires calibrated geometry, resolved phase dynamics, validated decoders and a spatial observable beside the resultant.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their

domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Unity has been approached differently elsewhere: integrated information theory treats integration and exclusion as constitutive (Tononi et al., 2016), global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene et al., 2014), and predictive-processing accounts derive both from a hierarchical generative model (Friston, 2010). Here the three requirements are kept apart rather than derived from one another, so that each carries its own assumptions and admits its own measurement; Section 9.4 returns to the comparison.

Consider reaching for a cup. A visual description of its location and a description of the hand’s relation to it can agree about where the cup is while contributing different contents. Synchronizing the two processes would leave disagreement about that shared quantity unresolved. An internal representation must additionally locate the perceiver within the represented situation. This example motivates the requirements; the mathematical constructions use spatial measures and fields in real normed vector spaces, whose identification with perceptual content is a further modelling task.

The mathematical contribution is to specify connections between these requirements and the assumptions each connection needs. A declared actuator’s stored energy bounds its mean coupling. Phase encoders shared on

overlaps convert phase order into a bound on content disagreement. Compatible spatial measures have a unique global extension; approximate agreement controls a specified global selection. A contracting reconstruction makes that extension its unique fixed point only if it accurately reconstructs the extension. Accuracy across a family of states imposes a further condition on the information retained by the encoding.

The empirical contribution is to identify what observations can decide. Stationary phase distributions can be shared by stronger coupling, weaker diffusion and a matched common drive. Finite ramps change the observable onset. A winding field is stationary at the same coupling as a uniform one and has the opposite resultant, so that observable does not determine the state. Coherent adaptation can lose alignment with a designated environmental template, and a compatibility estimator needs diagnostics that distinguish content from decoder artifacts. These findings determine the measurements needed to test the proposed connections.

Three of the ingredients are standard: the synchronization threshold $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988), the unique gluing of compatible local sections, and the Banach fixed-point theorem (Banach, 1922). Four things here are not. The installed-energy condition bounds mean coupling by the stored energy of a declared actuator, which makes a resource condition necessary for coherence on the stationary branch. The coherence-to-content bound converts phase order into a statement about descriptions rather than about timing, and its patch-local form is what keeps

58 it from being vacuous in a large population. A stability criterion on the
59 reweighted coupling kernel, together with an existence condition on an am-
60 plitude layer, settles when a low resultant reports weak coupling and when
61 it reports a winding field. And the composition is machine-checked together
62 with a witness discharging all eight hypotheses at once, which is what makes
63 their conjunction non-empty rather than a list no object satisfies.

64 Granting those premises, collected in Table 1, the account runs as follows.
65 Extracellular contraction on awakening raises endogenous field amplitude
66 and with it the effective coupling between cortical oscillators. Once that
67 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
68 within a quarter turn the population settles onto it; other initial data leave
69 windings stationary at the same coupling. Encoders shared where cortical
70 territories overlap turn that phase order into agreement about the quantities
71 the territories describe in common, so the local descriptions glue: there is
72 one state of which each is a restriction. Part of that state encodes the state
73 itself, and a readout of that part reconstructs it accurately enough, and
74 sensitively enough across the states that matter, to make it the unique state
75 the system models itself as being. On this account the glued state is what
76 makes distributed activity one situation, and its accurate self-reconstruction
77 is what gives that situation a perspective. Every step of it is conditional.

78 Section 2 collects the results that hold whatever physical field is proposed
79 and whatever substrate carries it; a reader who rejects the cortical hypothesis
80 of Section 8 still has to accommodate them. Sections 3–5 then develop the

81 three requirements together (Fig. 1). Section 6 composes them, stating the
 82 thermodynamic constraints and all eight connecting hypotheses per branch;
 83 its supporting models of feedback, learning and resource supply have distinct
 84 state spaces and assumptions. Section 7 then examines what the observables
 85 can establish, before Section 8 develops the cortical hypothesis and an awak-
 86 ening experiment. The hypothesis is that extracellular contraction increases
 87 field-mediated coupling; its direction, rate and relation to recovery require
 88 independent calibration. Supplemental Material organizes the constructions
 89 and controls behind the claims they support.

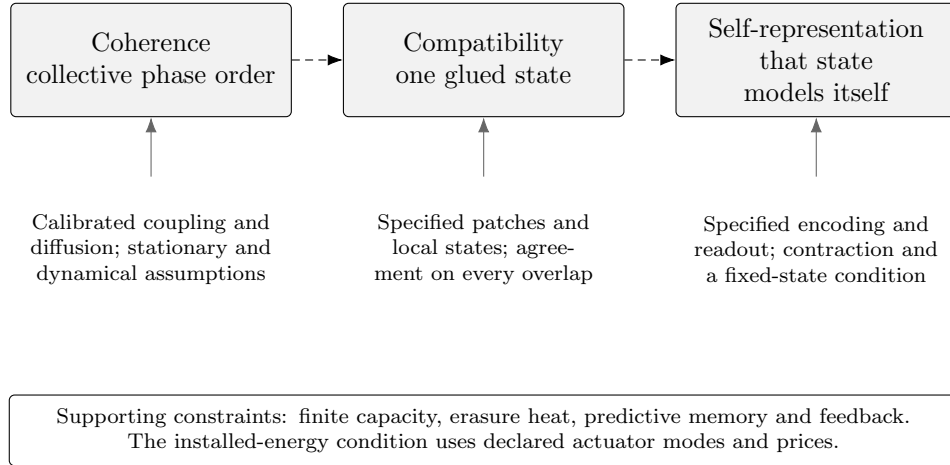


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

90 2 What holds without the cortical hypothesis

91 Most of what follows is conditional. Table 1 names eight hypotheses that
92 the composition consumes rather than derives, and the cortical identification
93 of Section 8 is a further commitment on top of them. Five results are not
94 conditional in that way. They constrain any account that reads unity off
95 phase coherence, whatever physical field is proposed to carry the coupling
96 and whatever substrate carries the field. Each is stated below where it is
97 used; collected here, they are the article’s fixed points.

98 **Phase order alone does not constrain content.** Assign every patch
99 the same phase and let two local mass profiles disagree at a shared site.
100 The profiles are chosen independently of phase, so synchrony bounds their
101 disagreement not at all, and they have no common extension: perfect locking
102 is consistent with local descriptions that do not glue (Section 4). What the
103 positive result of Section 4.2 costs is exactly this gap. A bound on content
104 disagreement needs encoders already related on overlaps, and that relation
105 is a supplied fact about the descriptions rather than a consequence of the
106 dynamics.

107 **The order parameter does not determine the state.** A field winding
108 once across a periodic sheet has $r = 0$ while every patch of it stays locked,
109 and this is not a contrived initial condition. On a periodic array of any
110 dimension, a winding is stationary for *any* isotropic symmetric kernel at

111 identical frequencies, because the coupling term is odd in the separation
 112 against a kernel even in it (`char_is_kuramoto_trajectory`). Section 7.3
 113 exhibits a sheet on which uniform and winding initial conditions reach 0.9657
 114 and 0.0083 at one coupling, diffusion and kernel. A low resultant therefore
 115 reports either weak coupling or a winding field, and no measurement in r
 116 alone separates them.

117 **A unique fixed point is not a self-model.** A constant readout con-
 118 tracts, fixes whatever state it returns, and reconstructs nothing. Uniqueness
 119 is consequently the wrong criterion, and Section 5.2 gives the right shape
 120 of one: reconstructing M states pairwise more than 2ε apart within error ε
 121 requires M distinguishable codes, so an alphabet of 2^b codes caps the family
 122 at $M \leq 2^b$. This is a counting argument and holds for any encoder and
 123 readout whatever.

124 **Where the encoding is a restriction, the region must resolve the**
 125 **family.** Read the code as the state's own restriction to a region and
 126 the criterion moves off the readout entirely. A boundary whose region
 127 reads two relevant states identically cannot reconstruct them more than 2ε
 128 apart (`not_isRestrictionResonance_of_reconstructs`), and a resonant
 129 boundary that does reconstruct a separated family restricts injectively on
 130 it (`restrictToAvatar_injOn_of_isRestrictionResonance`). Where the
 131 region sits is fixed before an encoding is chosen, so no readout supplies
 132 resolution the region lacks.

133 **Agreement after a change takes communication.** In a synchronous
134 network whose updates read only the site’s own state and its declared neigh-
135 bours’ messages, two configurations agreeing on a site’s causal past through
136 T rounds leave that site identical at round T . Two interventions outside
137 that past that require reports more than 2ε apart therefore cannot both be
138 reported from it within ε by the deadline (Section 5). Snapshot compatibil-
139 ity and timely compatibility are therefore different requirements, and cutting
140 the path removes the second while leaving the first intact.

141 None of these depends on the coupling being electromagnetic, on the
142 substrate being cortex, or on the eight hypotheses of Table 1 holding. Sec-
143 tion 9.3 applies the last three to a candidate that is neither biological nor
144 field-coupled.

145 **3 Coherence: physical coupling and its re-** 146 **source condition**

147 **3.1 Dynamics and the stationary threshold**

148 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

149 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
 150 K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field
 151 coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct
 152 parameters. The stationary calculation below uses identical frequencies and
 153 this scalar coupling. Applying it to a spatial kernel requires an aggregation
 154 rule and a check of the resulting approximation.

155 For identical frequencies in their rotating frame, the positive normalized
 156 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

157 where I_n are modified Bessel functions and ψ is the collective phase. The for-
 158 mal derivation uses the absence of stationary current for a periodic gradient
 159 drift to obtain this density, then verifies normalization and self-consistency.
 160 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above
 161 threshold, exactly one positive solution coexists with zero and increases
 162 strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

163 Figure 2 shows this stationary branch.

164 Linearization about the uniform density gives rate $K/2 - D$ for the first
 165 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 166 exactly above threshold. About the coherent density, the quadratic form

167 is strictly dissipative transverse to collective rotation, whose rate is zero.
168 These linear estimates concern the limiting equation. Relating them to finite
169 stochastic trajectories requires a dynamical mean-field limit and branch selec-
170 tion; exchange symmetry alone does not establish the needed independence
171 (Sznitman, 1991).

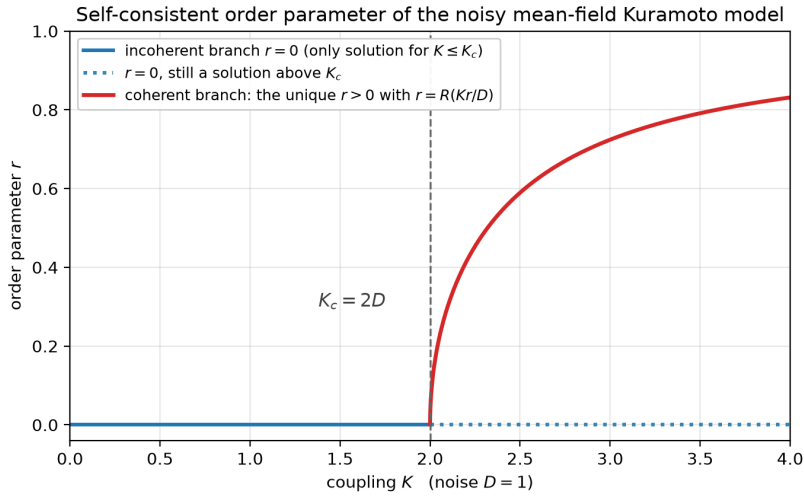


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

172 3.2 Installed energy bounds mean coupling

173 A spatial model replaces the finite sum by integration against a kernel. One
174 declared implementation carries response modes with continuous profiles ϕ_i ,

175 installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

176 The same configuration sets the reciprocal kernel and stored energy. An
 177 executed transition changes the kernel through its changed occupancies and
 178 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
 179 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 180 pair reconstruction on refining spatial partitions converges to its product-
 181 measure integral under the specified compactness and sampling assumptions;
 182 the refinement index is independent of the number of control updates.

183 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 184 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 185 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_z p(z) U(z), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \quad (5)$$

186 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 187 coherent stationary branch and any growing Fourier mode about the uniform
 188 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 189 is necessary: energy above it need not produce sufficient coupling because
 190 the bound can be loose.

191 The quantity constrained is installed energy rather than dissipated heat,
 192 and how much the condition excludes depends entirely on κ , which the mode
 193 profiles, prices and substrate normalization supply. Any macroscopic sub-
 194 strate carries installed energy far above the thermal floor, so the inequality
 195 binds through that conversion and not through the energy, and no result here
 196 derives κ for cortex or for any other candidate; what the condition contributes
 197 is a resource requirement in units a measurement can reach. Applying the
 198 scalar threshold to heterogeneous or signed spatial dynamics requires a jus-
 199 tified reduction, and identifying the installed kernel with a cortical field is a
 200 separate physical commitment.

201 **4 Compatibility: from coordinated activity** 202 **to a global description**

203 **4.1 Why common timing is insufficient**

204 For overlapping patches U and V , compatibility means that restricting either
 205 local description to $U \cap V$ gives the same object. Descriptions can agree on
 206 shared quantities while contributing different information elsewhere. Com-
 207 patible sections of the specified sheaf have a unique global extension. On
 208 a finite cover, local finite spatial measures agreeing as measures on every
 209 overlap also have a unique finite-measure extension whose sheaf image is the
 210 glued section. No mass-one normalization is imposed.

211 In the cup example, visual and reaching descriptions must express a
 212 shared quantity, such as cup location, in common coordinates. Restriction
 213 retains that quantity while dropping details particular to each description.
 214 Agreement permits surface appearance and hand configuration to contribute
 215 to one situation. The mathematical counterexample assigns the same phase
 216 to every patch while two local mass profiles disagree on a shared site. Those
 217 profiles have no common extension. Because they are chosen independently
 218 of phase, synchrony places no constraint on their disagreement.

219 This distinction limits the biological interpretation. Receptive territories
 220 and content domains must be related to spatial patches, and the descriptions
 221 must support the required extension. Matching overlap marginals of de-
 222 coded probability laws need not determine a unique global joint distribution
 223 (Abramsky & Brandenburger, 2011).

224 The cover is itself a choice, and the construction constrains it in one way
 225 only: synchronized patches agree where they overlap. Where they do not
 226 overlap there is nothing to compare, so a cover by pairwise disjoint patches
 227 satisfies that obligation for *every* assignment of local descriptions and phases
 228 (`ofDisjointCover`) and the gluing theorem still returns a unique global sec-
 229 tion, which is the tuple it was handed (`glue_unique_of_disjoint`). Member-
 230 ship of the class then reports nothing about the system beyond the patches
 231 covering it. What a gluing claim asks of a cover is therefore a non-degeneracy
 232 the theorem does not assume: that distinct patches meet (`HasNonempty-`
 233 `Overlaps`), and that what they share is a quantity fixed independently of

234 the agreement it is being used to establish.

235 4.2 When coherence does constrain content

236 A declared relation between phase and content gives a positive result. Let
 237 contents take values in a real normed vector space E . Patch i reads an
 238 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 239 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 240 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 241 sites; their equality on overlaps is a supplied relation between descriptions of
 242 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 243 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 244 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

245 Here discrepancy is the norm of the difference, and r and N are read on
 246 whichever population the bound is applied to. That choice is what decides
 247 whether the estimate says anything. A patch carries an order parameter of
 248 its own, and reading the bound there carries that patch's resultant and that
 249 patch's site count (`chord_le_of_patch_coherence`, `compatible_of_patch_`
 250 `coherence`), so a large population observed through small patches is con-
 251 strained by the small count. Read instead on the whole array, the factor N
 252 makes the estimate weak, because it draws a pointwise guarantee from an

average over N^2 pairs. At $r = 1$ the observations agree exactly at either
scale. A shared scalar encoder is a special case. Dropping the overlap rela-
tion permits disagreement even at perfect locking. A common nonexpansive
linear projection preserves agreement, but agreement of a projection does
not establish agreement of the full vector.

What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
constrains anything only where $\sqrt{2} N \sqrt{1 - r^2}$ falls below that. The winding
sweep of Section 7.3 fixes both ends of that range on one sheet. Read on the
sweep's own patch, 625 sites at patch-local order 0.9015, the bound is $382.5 L$
and constrains nothing; holding that order fixed, the count alone brings it
below the maximum chord only at 3 sites or fewer. A nearest-neighbour pair
of the same winding, the smallest patch there is, gives $0.1388 L$. Shrinking
patches therefore thins the overlaps at the same rate as it tightens the bound,
and the two ends are not reached by the same cover. At a columnar grain
the estimate constrains agreement inside one locally synchronous population
and is vacuous across the distant territories whose agreement unity is about.

Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-

274 duce an initial mismatch. Approximate observations leave a nondecaying
 275 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 276 update to the phase process.

277 4.3 What approximate agreement determines

278 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 279 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 280 Weights that vanish outside their patches and sum to one at each site select
 281 a global profile by weighted averaging. The selection lies within ε of each
 282 patch there, and any two such selections differ by at most ε , independently
 283 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 284 $L\varepsilon$ across selections.

285 At zero tolerance, the selection is the unique exact extension. At every
 286 positive tolerance there are families with no exact extension; selection then
 287 gives an approximation whose dependence on weights is bounded. Supple-
 288 mental Material gives a three-site witness attaining the bound. The mass-
 289 profile construction uses the scalar case of the content dynamics and addi-
 290 tionally requires nonnegativity of the initial profiles and observations. The
 291 general vector result supplies no identification with the spatial-measure sheaf.

292 These constructions concern declared objects: a specified cover, contents
 293 valued in a normed space, and encoders related on overlaps. They select
 294 neither the neural cover nor the decoder. Identifying their contents with
 295 perceptual content, and establishing that the identification bears on experi-

ence, is a further modelling task; Section 7 sets out what a measurement of
it would have to establish.

5 Self-representation: a state that models it- self

5.1 The reflexive construction

Compatibility gives a global state. Reflexivity relates it to an encoding within
that state. Equip global sections with a complete metric. If the additional
restriction-resonance condition identifies encoding with restriction to a region
 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (8)$$

The region and readout are fixed across the state space. If F contracts and
fixes the section s_* formed from the local descriptions, Banach’s theorem
makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
one for $\tau > 0$. Constructing a map with that factor remains a modelling
obligation.

In the cup example, the global state includes the represented scene and the

313 perceiver's relation to it. A local encoding must retain enough information
 314 for the readout to recover that modelled state; cup position alone would be
 315 insufficient if hand configuration and viewpoint could not be reconstructed.

316 For the same map with Lipschitz factor $q < 1$, approximate reconstruction
 317 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (9)$$

318 At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for
 319 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 320 weakens.

321 5.2 Reconstruction across a family of states

322 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 323 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 324 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

325 Thus M relevant states pairwise more than 2ε apart require at least M
 326 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 327 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 328 existing self-prediction map.

329 This is an additional accuracy criterion on a family, with no requirement
 330 for multiple exact fixed points. A blind encoder or constant readout fails it

331 for separated states despite having a unique fixed point.

332 Where the encoding is read as the state’s own restriction to a region, the
333 criterion becomes a condition on the region. Restriction resonance requires
334 the encoder and that restriction to identify the same pairs of states, so a
335 boundary reconstructing two relevant states more than 2ε apart while its re-
336 gion reads them identically is not resonant (`not_isRestrictionResonance_`
337 `of_reconstructs`); read forwards, a resonant boundary that reconstructs
338 a pairwise separated family restricts injectively on it (`restrictToAvatar_`
339 `injOn_of_isRestrictionResonance`). Which states a region reads identi-
340 cally is fixed by where the region sits, before an encoding is chosen, so the
341 resolution the family demands is the region’s and no readout supplies it. The
342 implication runs one way: a region that does resolve the family is not thereby
343 shown to carry a resonant encoding, and E89 supplies no such resolution.

344 **5.3 Agreement after a change takes communication**

345 Snapshot compatibility does not specify how quickly descriptions respond
346 when a shared quantity changes. In a synchronous network whose updates
347 read only the site’s own state and declared neighbours’ messages, two initial
348 configurations agreeing on a site’s causal past through T rounds produce the
349 same state there at round T . If two fresh interventions outside that past
350 require reports separated by more than 2ε , no report from that site can be
351 within ε of both. This follows from the reconstruction bound in Section 5.2.

352 A three-site delay line witnesses failure after one round and exact report-

ing after two; cutting the path prevents delivery at every round. A report
may still happen to be correct about one fixed world. The bound concerns
responsiveness across interventions.

Throughout this section *prediction* names a reconstruction map; reading
it as temporal neural prediction would require dynamics. The relevant family
of states, the metric, the tolerance, the communication graph and the dead-
line are all declared inputs, as are the represented variables and the fidelity
counted as accurate. Reading the resulting code count as physical memory
capacity requires a hardware model, and a readout that varies with the state
requires a further one.

6 Thermodynamics and composition

The phase and representation results concern coupling, content disagreement
and reconstruction error. Thermodynamics constrains the physical opera-
tions that could implement them, and it does so from the side rather than
along the spine: no edge below derives coherence, compatibility or reconstruc-
tion from a heat bound, and the resource condition these edges terminate in
is necessary without being restrictive for any macroscopic candidate (Sec-
tion 3.2). What this half contributes is a ledger in which a resource claim
is stated in checkable units. Finite-register capacity obeys $H(p) \leq \log |X|$;
erasure heat depends on specified system–bath assumptions (Landauer, 1961;
Norton, 2005; Shenker, 2000). A separate boundary result concerns a field

374 reaching disconnected vacuum components (Kibble, 1976). The composition
 375 connects these objects through the explicit hypotheses in Table 1.

376 **6.1 Feedback changes the information constraint**

377 For passive memory X_t driven by an environmental signal S_t , the predictive
 378 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

379 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
 380 et al., 2012). Zero retained information also satisfies it, so a task and a
 381 learning rule require additional specification. Relating this bound to Lan-
 382 dauer heat requires a common thermal scale and budget, and memory waste
 383 bounded by erased entropy.

384 An acting agent changes the signal it will receive. Specify policy, world,
 385 observation and memory-update channels. For their finite laws,

$$\begin{aligned} I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\ &\quad - I(X_t; S_{t+1} \mid S_t). \end{aligned} \quad (12)$$

386 The second conditional term vanishes in the passive model. With feedback,
 387 the signed difference can be negative as action creates correlations with the
 388 future. A causal reading requires an environmental state sufficient to describe
 389 the relevant dynamics.

390 For a controlled substep, the active thermodynamic bound instead uses

391 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F||P_R) =$
 392 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
 393 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
 394 2013). Every forward path of positive probability must have a positive-
 395 probability reverse for this finite identity. A missing reverse gives infinite
 396 divergence. The supplement develops the support conditions and composes
 397 actuation with memory updates on their shared intermediate law.

398 **6.2 Learning and repeated operation need explicit re-** 399 **sources**

400 The finite examples distinguish selecting a feasible policy, relaxing a register
 401 toward an encoded reward, and learning an unknown task from observa-
 402 tions. The observational learner acquires correlation with an environmental
 403 parameter while its action marginal stays uniform. A continuing model exe-
 404 cutes actuation, observation-driven learning and reset in sequence, preserving
 405 above-chance performance at a positive work cost per cycle.

406 A phase-derived sensor makes the informational distinction concrete: an
 407 informative readout and a blind control have the same phase order but dif-
 408 ferent learning performance. Attaching the learner’s register to the actuator
 409 of Section 3.2 ties reward, installed kernel and work to its own executed law.
 410 The learning witness rises above the necessary installed-energy bound while
 411 its matched control remains on the exclusion line. These finite constructions,

412 detailed in Supplemental Material, leave the evolving feedback between phase
 413 dynamics, installation and sensing to be specified.

414 For that model, common energy E , thermal scale θ , local detailed balance
 415 and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (13)$$

416 where expected cumulative work stays within the supplied allowance B at
 417 every prefix. This bounds expectations. Separate constructions track a store
 418 on each trajectory, include finite sources, charge sensor clearing and prepend
 419 preparation. They specify when a mean allowance represents an available
 420 resource; an expected budget alone permits overdrawn paths.

421 These constructions establish costs for their declared processes. A finite
 422 source with a positive net cost permits only finitely many funded cycles,
 423 while continuing operation requires replenishment. Register–bath witnesses
 424 also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 3.2 requires physical relations that these separate models do not supply
 425 automatically.
 426
 427

428 **6.3 What the composition establishes**

429 The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named
 430

431 substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

432 with B the register’s heat. The upper allocation follows from a ledger whose
 433 operations deliver that total heat to one reservoir and whose other operations
 434 do not increase joint entropy. Those ledger assumptions are physical inputs.
 435 Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.

Table 1 (continued)

Connection	Status	Required assumption
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours.
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate

441 resonance condition of Eq. (8).

442 Each of the eight edges is supplied rather than derived from the one above
443 it. E12 is equivalent to its conclusion, so capacity and boundary are inde-
444 pendent roots rather than a step from one to the other; both E45 discharges
445 take their convergence from refinement data and not from the thermody-
446 namic bound; E78 states that coherence relaxes the supplied cover, and the
447 coherence-to-content estimate of Eq. (6) bounds disagreement only for en-
448 coders already shared on overlaps. The inferential content of the composition
449 is its last step, where contraction and a fixed glued section give uniqueness.
450 What composing them establishes is that the premises are consistent, named
451 in one place and machine-checked together, and not that coherence follows
452 from installed energy or self-representation from coherence.

453 Joint witnesses combine a register, boundary, refining mesh and small
454 spatial cover; active witnesses add controlled thermal updates and a priced
455 installation. They establish satisfiability under their stated identifications.
456 Mesh convergence follows from refinement data, and the witnesses establish
457 no common cortical mechanism. Supplemental Material maps the claims to
458 their Lean declarations and records the counterexamples. The development
459 declares no additional axioms beyond its libraries' foundational axioms.

460 7 What the observations establish

461 The composition holds under its premises. Whether observation can estab-
462 lish those premises is a separate question, and the three requirements it
463 connects concern different objects: phase dynamics, local descriptions and
464 internal encodings. Their observables must retain these distinctions. The
465 following controls show which mechanisms remain indistinguishable, which
466 temporal approximations fail, which states one observable cannot separate,
467 and what independent content measurements must establish. Each is a finite
468 numerical regime on a declared model, with full parameters and controls in
469 Supplemental Material, and none of them calibrates a cortical quantity; the
470 scope notes below concern what each result does and does not separate.

471 7.1 Stationary distributions leave the mechanism un- 472 determined

473 Equation (2) gives a concentration–coherence relation with no fitted shape
474 parameter. Estimate r from circular resultant length and a from density
475 shape: inverting R from the measured r would make agreement automatic.
476 Even independently estimated agreement identifies a statistical family shared
477 by different mechanisms.

478 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
479 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
480 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled

481 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t) r(q(t))$,
 482 and diffusion D_0 have that same density. Their stationary distributions co-
 483 incide window by window despite different causes.

484 Resolved dynamics supply additional information. In the ideal model with
 485 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
 486 increment means constrain drift. Separating endogenous coupling from a
 487 matched common drive additionally requires a measured drive or controlled
 488 perturbation; identifying electromagnetic coupling requires evidence specific
 489 to that interaction. Supplemental Material gives the equivalence construction
 490 and its numerical check.

491 **7.2 Finite ramps change the observable onset**

492 The square-root law in Eq. (3) describes stationary solutions. A changing
 493 coupling requires time for the phase distribution to approach those solutions.
 494 In a deterministic mean-field control that crosses the threshold symmetrically,
 495 none of the scanned rates meets the declared tracking tolerance: slowing
 496 the crossing does not remove the terminal residual under that initialization.
 497 A slow control variable therefore supplies no general guarantee of branch
 498 tracking at onset. Supplemental Material gives the residual, rate range and
 499 admissible rates away from threshold.

500 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
 501 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
 502 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay

503 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
 504 6 uncensored legs supply the delay scaling below; the decade-spaced ones
 505 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
 506 onset exponents differ from those of the exact stationary branch evaluated
 507 on the same grids. At the fastest speed the ensemble mean never reaches the
 508 fit window, so its fitted slope does not estimate stationary onset.

509 Delayed passage through a noisy bifurcation (Berghlund & Gentz, 2002)
 510 makes the apparent exponent depend on ramp speed and fitting window.
 511 The specified deviation-based critical ramp speed is not bracketed by this
 512 sweep. With fluctuating coupling, the relevant stationary comparison also
 513 depends on rate: the mean coupling approximates order in the tested fast
 514 regime, while averaging stationary order over coupling works for the tested
 515 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 516 neither approximation.

517 **The prediction.** The delay scaling is what this section supports as a pre-
 518 diction, and it needs no absolute calibration. Over the 6 uncensored legs the
 519 mean delay follows $\Delta K \propto v^{0.444}$, with a 95% slope interval of [0.394, 0.494].
 520 The reference for that exponent is the same estimator applied to a deter-
 521 ministic mean-field run seeded at the fluctuation floor, which returns 0.425
 522 over the same legs at the same escape criterion, inside the interval on the
 523 measurement. The distance from 1/2 is what an absolute escape criterion
 524 costs: the run crosses the saturation between the floor it starts from and the

level it is stopped at, and tightening the criterion to $r \geq 0.05$ returns 0.490.
 Stated as a law rather than as a number: onset delay scales as the square
 root of the rate at which coupling crosses threshold. Testing it requires only
 that $\dot{K}$ be monotonic in a manipulated rate — not its value, nor κ , nor γ
 — so it is a within-subject design that varies the rate of emergence from
 anaesthesia alone, with each participant as their own control. Such a design
 must avoid the hazard this sweep exhibits: a fixed absolute escape criterion
 meets a size-dependent critical fluctuation floor, so the criterion has to be
 referred to each recording’s own pre-threshold fluctuation scale rather than
 to a shared number. The diffusion D is the remaining input, and Section 7.1
 notes that it is estimable from resolved phase traces through the rate $2D$
 at which quadratic variation accumulates, which is a different quantity from
 the concentration the propofol exercise could not range over.

Additional controls address integration bias, population size and spatial
 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 $[1.958, 2.009]$ contains 2.

7.3 A winding state leaves the order parameter ambiguous

The order parameter r is indifferent to where a phase sits in space. A field
 winding once across a periodic sheet therefore has $r = 0$ while every patch
 of it stays locked. Such a field is not an artificial construction here. On a

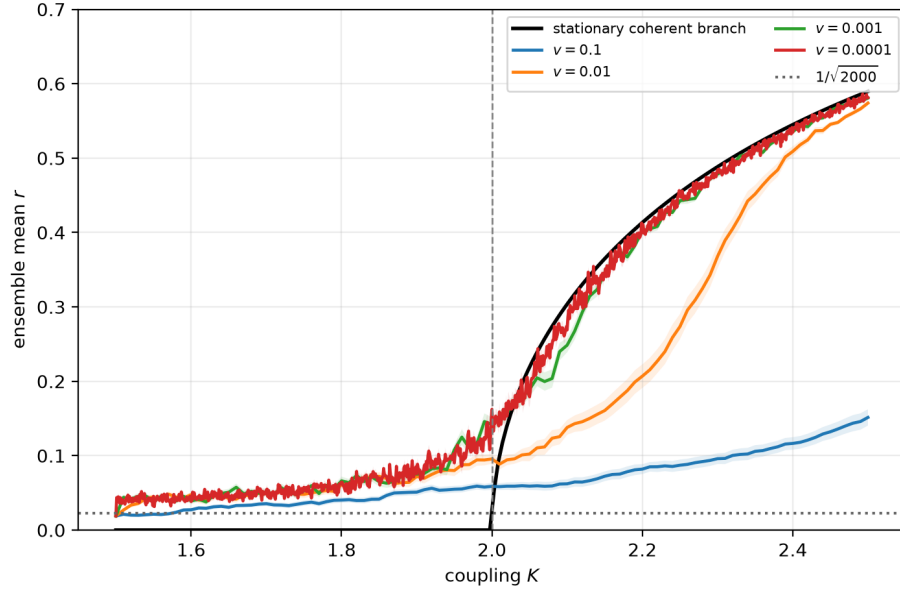


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

546 torus a winding is stationary for any isotropic symmetric kernel at identical
547 frequencies: the coupling term is odd in the separation against a kernel even
548 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves
549 this for any finite periodic array, this sheet included. A winding satisfies the
550 positivity and identical-frequency hypotheses. It violates only the quarter-
551 turn initial spread that the convergence result additionally requires before
552 placing the limit at the uniform state.

553 On the sheet of Section 3's spatial control, a winding survives below
554 an interpolated decay length of 0.1871 mm and unwinds above it. That

555 boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the
 556 same coupling, diffusion and kernel reaches $r = 0.9657$ from a uniform start
 557 and $r = 0.0083$ from a winding one. Both are stationary, with patch-local
 558 order 0.9015 and no winding defect. At fixed K the observable therefore takes
 559 two values spanning nearly its whole range. Coupling does not determine the
 560 resultant, and a low resultant does not by itself distinguish weak coupling
 561 from a winding field.

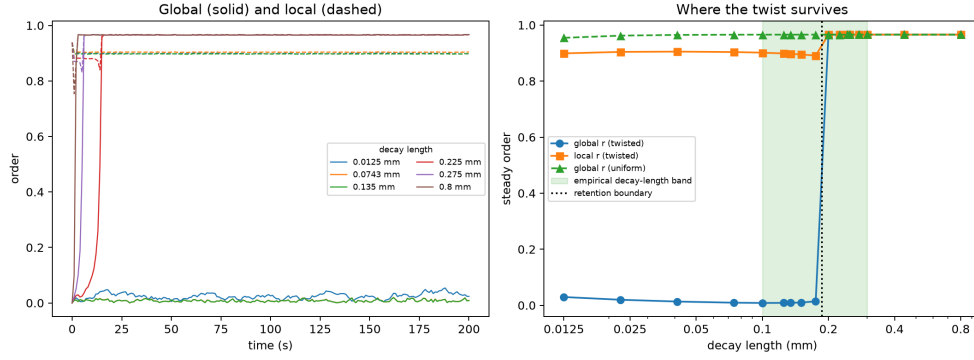


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

562 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-
 563 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 564 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.

565 Heterogeneity shortens the range that sustains a wave without removing it.

566 Which windings persist has a criterion behind it. Linearizing the phase
567 model about a winding returns a coupling of the same translation-invariant
568 form, each separation reweighted by the cosine of the phase the winding
569 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is
570 again even in the separation, so the same characters diagonalize it and ev-
571 ery mode's rate is one of its eigenvalues. Where the winding advances by
572 at most a quarter turn across every separation the coupling reaches, the
573 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
574 `stable_of_quarter_turn`); where some rate is positive, that mode grows
575 without bound (`charJacobian_growing_mode`). Moving coupling mass to
576 longer separations is what carries a phase advance past a quarter turn, and
577 that is the mechanism the decay-length sweep varies. The criterion bounds
578 rates rather than trajectories: asymptotic stability is not proved, so the
579 states that persist occupy a narrower range than it admits, and the bound-
580 ary quoted above is measured and not derived from it.

581 The same reweighted kernel measures the potential's curvature. A wind-
582 ing has $r = 0$ where a global minimizer of the Lyapunov potential has $r = 1$,
583 so it is not one (`char_not_potential_min`): the quarter-turn condition on ini-
584 tial data does not merely decline to cover a winding, it must exclude it. Along
585 a mode, the potential's second variation is that kernel's quadratic form, so a
586 growing mode is a direction of falling potential and a winding carrying one
587 is not a local minimum either (`char_not_local_min_of_charLambda_pos`).

588 An amplitude layer decides which of these states exist, and leaves what
 589 the resultant reports unchanged. A phase state carries no amplitude. Ev-
 590 ery oscillator sits on the unit circle, so a field with an amplitude node has
 591 no representation in these dynamics. The real Stuart–Landau extension
 592 supplies one on the same coupling, and leaves intact the gradient struc-
 593 ture the convergence results rest on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i =$
 594 $(\mu - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its phase equation is the Kuramoto equation
 595 with each coupling weighted by a ratio of amplitudes (`amplitude_phase_is_`
 596 `kuramoto`). The phase model is this one with the amplitude ratios set to
 597 one, rather than a separate object.

598 What the added degree of freedom decides is which windings exist at all.
 599 On a periodic array of any dimension, this sheet included, a winding requires
 600 $\mu + \lambda > 0$. Here $\lambda = \sum_d f_d(\cos \phi_d - 1)$ is the eigenvalue the kernel supplies,
 601 summed over separations, with ϕ_d the phase the winding advances across d
 602 (`exists_charState_iff`; on the ring $\phi_d = 2\pi qd/n$). For the nonnegative
 603 kernels used here $\lambda \leq 0$, and moving coupling mass to longer separations
 604 lowers it further. A broad enough kernel therefore removes the winding out-
 605 right. The sweep above reaches that same boundary by a different route:
 606 the phase model retains every winding at every coupling, and loses this one
 607 under integration instead. The two layers bound the same regime from op-
 608 posite sides. A phase recording does not separate a state the amplitude layer
 609 excludes from one the dynamics fails to sustain.

610 These states are windings and not spirals, and on the sheet that is a

theorem rather than a shortage of dimensions. A cortical phase singularity is a point at which the amplitude vanishes and around which the phase turns once, read in practice as the circulation around a small plaquette. The states above have equal modulus at every site, so that circulation is exactly zero on every plaquette (`plaquetteCirculation_eq_zero_of_winding`). Nor can the layer host a single defect. Over a periodic sheet the circulations of *any* phase field sum to zero, so defects arrive in cancelling pairs (`no_isolated_defect`). The sweep’s near-zero measured defect density agrees with the first of these and says nothing beyond it.

Two limits bound the layer for a reader of the cortical literature. Relaxational dynamics has no propagation speed, which is the quantity a cortical wave measurement leads with. And $|z|$ is a normal-form radius, which a measured envelope is not; this work identifies the two nowhere. Supplemental Material gives the sweep, its uniform control, the amplitude layer and the standing wave it admits inside the band, together with the remaining limits of both.

A vanishing resultant is the absence of global agreement and not of agreement. Eq. (6) read on a nearest-neighbour patch of a winding gives a discrepancy bound of $2\sqrt{2}L|\sin\phi|$, with ϕ the phase advanced across the patch, which falls to zero as the same winding is spread over more sites (`chord_le_of_char_patch`), while the global form of the same bound is vacuous at $r = 0$. Encoders shared on overlaps therefore still bound content disagreement inside a patch of a state that no global observable calls coherent. The winding

634 number itself stays unconnected to content.

635 This qualifies what a phase trace can establish. An onset criterion stated
636 in r alone registers a change of basin as readily as a change of coupling.
637 The protocol of Section 8.3 therefore needs a spatial observable beside the
638 resultant.

639 **7.4 Coherent adaptation can lose environmental align-** 640 **ment**

641 An adaptive coupling rule might supply representational structure, but its
642 objective must reward the relevant agreement. For symmetric coupling, pair-
643 wise sine terms cancel in the population mean drift. Minimizing squared drift
644 at that fixed mean therefore rewards equalizing drift across nodes. In the
645 frequency-cluster control, this favors coupling between unlike-frequency clus-
646 ters, whereas the diagnostic environmental template rewards coupling within
647 clusters.

648 The formalization proves descent of the squared-drift objective for a spec-
649 ified gradient flow at fixed phases. Its functional differs from thermodynamic
650 entropy production: at a coherent stationary density the probability current
651 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
652 rial gives the operator, its gradient and the conditions of the descent result.

653 The finite control co-evolves noisy phases and symmetric, nonnegative
654 couplings at fixed total resource. Its template is a diagnostic covariance of

latent signals, never a training input. Across three seeds, second-half order remains between 0.9167 and 0.9222, while the within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5). Random-update, permuted-gradient and frozen-kernel controls are reported in Supplemental Material.

The immediate post-update objective reduction disappears when averaged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

7.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on constructed data separates

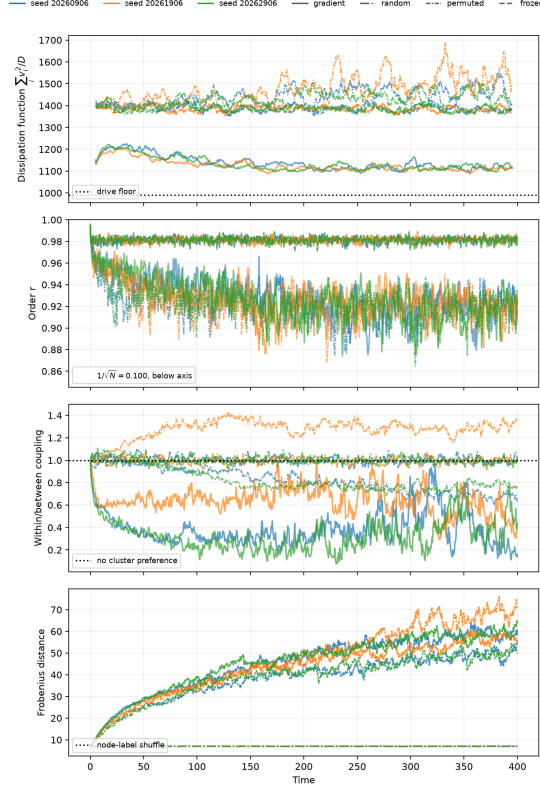


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

677 these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by
678 trial; the compatible construction scores 0.476 on an interval containing one
679 half. Shared-prior shrinkage can nevertheless manufacture agreement, and
680 regional bias can manufacture disagreement. Per-patch decoding diagnostics
681 detect these failures; the compatibility statistic alone does not.

682 Phase agreement cannot replace that measurement. On constructed con-
683 tent, phase carries a share 1.000 of the content entropy when content is built
684 from it and 0.000 when it is not. A phase-derived statistic reports com-
685 patibility on phase-locked patches whose local descriptions disagree. The
686 implemented content statistic detects their incompatibility. These controls
687 validate a measurement on known constructions; a neural content model and
688 decoder remain to be established.

689 Choosing the cover also chooses a candidate subject: trivial descriptions
690 can agree while carrying little content, and a partition can conceal disagree-
691 ment. The useful choice must therefore be constrained by what neural ac-
692 tivity encodes. The exact composition assumes compatibility of its supplied
693 cover. The content mechanisms above identify additional hypotheses under
694 which agreement can arise; they leave the empirical choice of content and
695 the connection to that cover explicit.

696 **7.6 What an informative phase-density measurement** 697 **requires**

698 The concentration range and the dependence structure decide whether the
699 concentration–coherence relation can be tested at all. Under the specified
700 log-density estimator and bias-floor criterion, discrimination at the observed
701 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
702 designs, rising to 31000 when sites share a mean in groups of 10. These
703 counts describe that estimator and dependence model; the intended recording
704 requires its own calibration. Supplemental Material reports the bin-count
705 controls and the full dependence-aware design.

706 That ceiling comes from exercising the estimators on public propofol-
707 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms windows.
708 Bipolar and common-reference montages produce different residuals. The
709 bipolar results are compatible with the Bessel relation over a concentration
710 range too narrow to distinguish it from its linear approximation at the ob-
711 served scatter, which is the range the design scan takes as given. Temporal
712 pooling changes the estimand, and sedation recordings leave the awakening
713 hypothesis untested.

714 An informative dataset therefore needs sufficient concentration range or
715 precision, calibrated estimators and an observation model for dependence,
716 source mixing and reference choice. Public mouse calcium imaging provides
717 thousands of spatially located neurons across waking, sleep and anesthe-

718 sia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-
719 estimation band used here. Site count, state coverage and bandwidth are
720 separate requirements.

721 **8 The cortical hypothesis and an awakening**

722 **test**

723 The cortical proposal identifies a physical source of coupling and asks whether
724 its changes help explain recovery. The tests in Section 7 determine what
725 observations must distinguish before a phase trace can support this identifi-
726 cation.

727 **8.1 Physical identification and calibration**

728 The cortical hypothesis identifies part of effective coupling with endogenous
729 extracellular electromagnetic interactions. Fields can modulate spike timing
730 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
731 propose coordination of distributed neural information (Pinotsis & Miller,
732 2022; McFadden, 2020). Phase organization under propofol motivates study-
733 ing this regime (Bastos et al., 2021), but does not measure coupling.

734 The formal field predicate requires a jointly continuous kernel, a normal-
735 ized substrate, coupling equal to the kernel average, and no single site pro-
736 viding more than half that strength. Toy constructions satisfy it. A cortical

737 realization must specify the scale at which the kernel and its reduction ap-
738 ply: calcium imaging reports different state-dependent modularity patterns
739 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

740 Cortical columns are the obvious candidate unit, and they answer one
741 of the two demands the formalism makes. As the oscillator index they are
742 a serious proposal: they are delineated by response properties and anatomy
743 rather than by the agreement they would be used to establish (Mountcastle,
744 1997), and a phase can be estimated from a column-sized territory at the
745 electrode spacings used for cortical wave measurements (Townsend et al.,
746 2015). As the content cover they are a poor one, for three separate reasons.
747 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
748 the compatibility obligation without constraining anything (Section 4). The
749 cover condition additionally requires the patches to exhaust the substrate,
750 while much of association cortex has no accepted columnar parcellation. And
751 the column’s standing as a canonical cortical unit is contested (Horton &
752 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
753 supplies the oscillator index and leaves the content cover to be identified on
754 separate evidence, which is the bridge assumption E78 records.

755 The spatial reduction is also testable. A row-sum aggregation at zero
756 diagonal tracks the uniform-coupling control over the tested spatial decay
757 range, while frequency heterogeneity displaces the operational threshold. The
758 fit window biases both estimates relative to $2D$; agreement with the control
759 is therefore the supported conclusion. Supplemental Material reports the

760 reduction, its numerical error and the distinct assumptions needed for the
 761 kernel-average identification in Eq. (5).

762 The field commitment requires physical evidence beyond a choice of math-
 763 ematical representation, and the available support comparisons do not supply
 764 it. A finite matrix spread over positive-mass cells preserves its total cou-
 765 pling and phase correlation exactly, while point-supported vanishing under
 766 an atomless measure and a disadvantage from excluding a better-correlated
 767 pair concern specified support constraints; Section 9.3 reads the same com-
 768 parisons against a digital candidate. Neither supplies energy prices.

769 A spike-time shift alone supplies no coupling rate. The dimensionless
 770 product of population size N , timing sensitivity s , field amplitude E and
 771 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (15)$$

772 This rate describes how induced timing shifts accumulate into drift. Nei-
 773 ther it nor the installation factor κ follows from a heat bound. Testing the
 774 physical threshold requires independent estimates of the interaction kernel,
 775 its aggregation and diffusion in the same cortical regime.

776 8.2 Extracellular geometry and recovery

777 The proposed control variable is extracellular geometry. Extracellular vol-
 778 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &

779 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
780 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
781 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
782 an effective-medium description predicts that a larger extracellular fraction
783 raises conductivity and reduces field amplitude.

784 The coupling-gated hypothesis consequently predicts that extracellular
785 contraction on awakening raises E and, holding the other factors in Eq. (15)
786 fixed, raises effective coupling. This argument supplies a direction; the
787 geometry–coupling relation, its magnitude and its role in human recovery
788 require measurement.

789 For a smooth crossing at nonzero speed, evaluating the stationary branch
790 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

791 This follows from Eq. (3) at fixed D and concerns the stationary branch
792 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
793 dently justified tracking approximation. The crossing controls in Section 7.2
794 leave that approximation unsupported at onset, even though a separation of
795 geometry and phase time scales can support it away from threshold. The
796 awakening hypothesis therefore needs a calibrated dynamical prediction for
797 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
798 does not supply a recovery latency.

799 8.3 A protocol for the awakening experiment

800 The physical test has six requirements:

- 801 1. **Coupling and diffusion in the same units.** Estimate them from re-
802 solved phase dynamics, separating heterogeneity and observation noise.
803 Measure the drive or perturb the system to distinguish endogenous cou-
804 pling.
- 805 2. **Geometry calibrated to coupling.** Validate the geometry observ-
806 able and its relation to effective coupling, with uncertainty. The con-
807 ductivity argument supplies only the expected direction.
- 808 3. **Calibrated phase measurements.** Estimate r by resultant length
809 and a by unweighted regression of log density on centered cosine.
810 Choose site count and binning from a dependence-aware calibration
811 over the intended concentration range. The tested designs require
812 thousands of sites at the observed concentrations; temporal pooling
813 changes the spatial estimand.
- 814 4. **An independent recovery endpoint.** Measure behavioral recovery
815 separately from phase organization.
- 816 5. **A predeclared dynamical onset criterion.** Set a coherence thresh-
817 old above the finite-size floor and evaluate its latency under the mea-
818 sured coupling trajectory, including ramp speed or fluctuations. Record

819 a spatial observable beside the resultant, since a winding field reports
820 a low resultant at coupling that sustains local order (Section 7.3).

821 **6. Matched model comparison.** Compare coupling increase, diffusion
822 change, common drive and alternative recovery models under the same
823 observation and filtering assumptions.

824 Failure of the concentration–coherence relation over an informative range
825 constrains the density model. Failure of the calibrated geometry–coupling
826 association constrains the physical gate. Better prediction of independent
827 recovery by a competing mechanism counts against its proposed role. A
828 favorable outcome still leaves the content of coordinated activity to be ex-
829 plained.

830 9 Discussion

831 Two things have been established here, and they are of different kinds. The
832 constraints of Section 2 hold outright: synchrony does not deliver compatible
833 content, the resultant does not determine the state, a unique fixed point is
834 not a reconstruction, a region that cannot resolve a family cannot encode it,
835 and agreement about a change takes communication. Any account that reads
836 unity off phase coherence has to meet them, this one included. The account
837 set out in Section 1 is conditional at every step by contrast: each connection
838 in it is a hypothesis supplied rather than a result derived, and Table 1 names
839 all eight. Four of the ingredients assembled there are not standard — the

840 installed-energy condition, which makes a resource requirement necessary for
 841 coherence on the stationary branch; the patch-local form of the coherence-
 842 to-content bound, which is what keeps it from being vacuous; the stability
 843 criterion and amplitude condition that settle when a low resultant reports a
 844 winding field rather than weak coupling; and a machine-checked composition
 845 carrying a witness that discharges all eight hypotheses at once, which is what
 846 makes their conjunction non-empty rather than a list no object satisfies.
 847 What follows takes the conditions in turn, as commitments the proposal
 848 incurs, as measurements that would bear on them, and as questions the
 849 framework leaves open.

850 **9.1 Unity and a minimal self as empirical commit-** 851 **ments**

852 We propose the state glued from compatible local descriptions as a corre-
 853 late of conscious unity, and its accurate internal reconstruction as a correlate
 854 of a minimal self. The motivations are that different contents belong to
 855 one experienced situation and that the situation is experienced from a per-
 856 spective. The conditional results identify relations a realization would need:
 857 sufficient installed resources for the specified phase model, agreement about
 858 shared content, and an encoding that supports reconstruction at the required
 859 fidelity. Discharging E12–E89 for cortex would make the composition a state-
 860 ment about cortex: that a cortical field above threshold relaxes to a cover

861 whose compatible local descriptions glue, and that the glued state is the
862 unique fixed point of a reconstruction map realized there. It would leave the
863 two identifications above exactly where they are, because neither is among
864 the eight and neither follows from their conjunction; they are the commit-
865 ment this subsection names, and Section 9.2 lists the three observations that
866 would count against it.

867 Compatibility allows different descriptions to contribute different infor-
868 mation. It also permits uncertainty and conflicting judgments: apparent cup
869 position and judged cup position can be distinct quantities in a consistent
870 description of their discrepancy. Which quantities are shared must be de-
871 termined independently. Choosing them solely to secure agreement would
872 deprive the proposal of empirical force.

873 The minimal-self proposal concerns the perceiver’s position and condition
874 within the situation, such as the cup’s location relative to a body that can
875 reach it. The encoding could be implicit and need carry no autobiograph-
876 ical narrative. Its proposed necessity is philosophical: even an unreflective
877 episode has a perspective. Gluing by itself requires no self-encoding, and re-
878 constructability alone supplies no functional role. Altering the encoded hand
879 or viewpoint while holding the external scene fixed should change estimates
880 of the perceiver’s relation to the cup as the readout predicts. Whether that
881 represented perspective is experienced remains a further question.

882 **9.2 What a test of the full proposal must establish**

883 Both uniqueness results concern specified objects. They choose neither the
884 cover nor the encoding, and do not select between candidate subjects. In-
885 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
886 here selection requires independent evidence about neural organization. A
887 constant readout can contract and fix a predetermined state while discard-
888 ing its input. The family criterion in Section 5.2 tests the sensitivity that
889 this construction lacks, provided the relevant states and fidelity are declared
890 independently.

891 A test must separately measure coupling relative to diffusion, content
892 compatibility and self-reconstruction, alongside an independent endpoint for
893 experience. Meeting all three without a unified episode would challenge suffi-
894 ciency. A unified episode with incompatible descriptions would challenge the
895 unity identification; one lacking the specified self-encoding would challenge
896 the reflexive requirement. Awakening, dreaming and dissociative states offer
897 possible settings, but absence of report cannot by itself define absence of
898 experience.

899 The next physical requirement is a calibrated geometry–coupling relation
900 and resolved phase dynamics. The installation bound contributes a nec-
901 essary resource condition where its hardware representation applies; cortical
902 mode prices and field-mediated coupling require measurement. For the repre-
903 sentational connection, an informative experiment asks whether a controlled
904 change in coordination alters independently decoded content agreement, with

905 controls for sensory information and decoder quality. The constructed-data
906 estimator establishes feasibility for specified contents. Neural local descrip-
907 tions, shared overlap encoders and the self-encoding have yet to be identified.
908 The population-size dependence of the phase-to-content bound further limits
909 its quantitative use, in the form Section 4.2 reads off the sheet.

910 The separation results explain why these measurements must be inde-
911 pendent. A stationary distribution leaves several recovery mechanisms indis-
912 tinguishable. Equal phase order can accompany incompatible descriptions
913 or different task information. The tested plasticity rule preserves coherence
914 while losing its designated template. Each finding concerns the specified
915 model and observable; their role is to delimit the evidence needed for the
916 proposed connections.

917 **9.3 What the conditions constrain in a digital candi-** 918 **date**

919 The results of Section 2 hold for any candidate, and for a large language
920 model (LLM) running on graphics processing units (GPUs) three of them are
921 checkable rather than merely stateable. That is where this framework bears
922 on claims made for such systems. It does not exclude them, and no result
923 here could; what it does is fix what a claim of unity or self-representation
924 commits its maker to measuring.

925 **Codes, not reports.** Reconstruction across a family is a counting con-
926 straint before it is anything else. If two macrostates the candidate is said
927 to represent satisfy $E(s) = E(t)$ while $d(s, t) > 2\varepsilon$, Eq. (10) rules out re-
928 constructing both within ε , whatever the readout. A system’s reports are
929 readouts. Identical verbal reports across two conditions are therefore consis-
930 tent with identical internal codes and settle nothing about the states, while
931 the converse claim — that the system represents its own condition across a
932 declared family — asserts that its internal code separates that family, which
933 is a statement about activations under intervention and not about text. The
934 measurement follows from the statement: declare the family and the fidelity
935 first, then test whether the encoding region distinguishes its members.

936 **The region, not the readout.** Where the encoding is read as the state’s
937 restriction to a region of the machine, the demand falls on the region before
938 any readout exists. For a decoder-only transformer the causal mask makes
939 the network an instance of the update model of Section 5: the activation at
940 position t and layer ℓ reads positions at most t at layer $\ell - 1$ and nothing
941 else, so a region’s causal past is fixed by the architecture rather than chosen
942 by interpretation. Two interventions outside that past that demand reports
943 more than 2ε apart cannot both be answered within ε from that region.
944 A claim that a model reports on its own processing consequently carries an
945 obligation to say which positions and layers the report’s causal past contains,
946 and a report that appears to track an intervention outside it indicates a

947 confound or a different region, not a faster channel.

948 **Agreement about what.** Unity requires the candidate’s local descriptions
949 to agree on independently identified shared quantities (Section 4). Common
950 timing, a shared residual stream and fluent output supply none of that, for
951 the reason Section 2 gives: coordination bounds content disagreement only
952 through encoders already related on overlaps, and which quantities two parts
953 of a model describe in common is a decoding question carrying the failure
954 modes of Section 7. The cover is a choice, and choosing it to secure agreement
955 empties the claim.

956 **What is not excluded.** Digital representations can satisfy compatibility,
957 so finite hardware is compatible with the gluing construction and none of
958 the above is a no-go for machines. The coupling-gated route E56–E78 is a
959 different matter: it requires a calibrated physical coupling, its noise regime
960 and a connection from coordination to content agreement, and an attention
961 matrix or a shared clock supplies none of these identifications. Comput-
962 ing a field model likewise leaves its realization in the computing substrate
963 to be established. Those physical obligations fall on the cortical proposal
964 identically.

965 The installed-energy condition of Section 3.2 is restricted in the same
966 way. It constrains a macroscopic candidate only through κ , which is hard-
967 ware data supplied by mode profiles and prices, and Supplemental Material
968 reads the condition against a cortical metabolic budget to return the conver-

969 sion factor it would require rather than a verdict. The hardware comparisons
970 there have similarly restricted scope: fixed wiring can impose a correlation
971 disadvantage, and point-supported kernels vanish under an atomless contin-
972 uum measure, but neither comparison makes the representation conditions
973 unattainable on GPUs. A universal exclusion would need a necessary condi-
974 tion for consciousness and a proof that every relevant implementation violates
975 it, and the present composition supplies neither.

976 Whether experience requires a living substrate is the explicit subject of
977 biological naturalism (Seth, 2026a,b). On that view the properties ground-
978 ing experience are those of self-sustaining organisms, and simulating them
979 does not secure them. The conditions stated here name no biological mech-
980 anism, but computation described as such does not satisfy them either: a
981 candidate must exhibit a physically realized coupling in a measured noise
982 regime, agreement about independently identified shared quantities, and an
983 internal reconstruction meeting a declared fidelity. Whether a realization
984 of those conditions that is not alive would be accompanied by experience
985 is the point at issue, and these results do not settle it. A related position
986 locates the difference in the mode of computation rather than in life as such:
987 biological systems compute across scales that do not separate, and along-
988 side discrete operations carry continuous-valued dynamics belonging to their
989 physical medium (Milinkovic & Aru, 2026). The coupling condition stated
990 here is one measurable instance of that requirement, fixed to a physical field
991 in a declared noise regime, and a description of such a field does not satisfy

992 it.

993 The neuroscientific case against *human-like* consciousness in text-only
994 LLMs concerns embodied sensory interaction, thalamocortical organization
995 and biological regulation linking perception and action to survival (Aru et al.,
996 2023). Their absence supports skepticism, conditional on their causal rele-
997 vance to experience. Those mechanisms are not requirements of the abstract
998 gluing and reconstruction definitions, and workspace-like representations re-
999 ported in some LLMs (Gurnee et al., 2026) supply candidate organization to
1000 investigate. What this framework adds to that investigation is an order for
1001 the questions: declare the family of states and the fidelity, fix the encoding
1002 region and read its causal past off the architecture, and only then ask what
1003 the system reconstructs. A claim that survives those three steps is a claim
1004 about the machine; one that does not survive them was a claim about its
1005 output.

1006 9.4 Explanatory scope

1007 Integrated-information, global-workspace and predictive-processing accounts
1008 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
1009 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
1010 candidate physical interaction and conditional relations between local content
1011 and self-encoding. Gluing and reconstruction permit multiple substrates;
1012 the electromagnetic commitment enters through physical calibration. The
1013 formalization checks implications under their hypotheses, and its witnesses

1014 establish realizability on declared models. Two things stand outside that
1015 result and are not supplied by it: the eight hypotheses are discharged for
1016 toy components rather than for cortex, and the identification of the glued
1017 state with unity and of its reconstruction with a minimal self is a further
1018 commitment that the eight do not contain.

1019 The explanatory question is whether the specified relations occur in neural
1020 activity and matter for experience. In the cup example, this means establish-
1021 ing how coordination helps maintain one represented situation and how the
1022 perceiver’s place within it contributes to the episode. The proposal identi-
1023 fies candidate correlates and tests of that identification. Why those relations
1024 would be accompanied by experience remains unexplained.

1025 **Data and code availability**

1026 The Lean development, simulation scripts, tests and saved numerical sum-
1027 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
1028 `cortical_coherence`. Publication simulation values are generated from
1029 saved summaries without rerunning production sweeps. The exploratory
1030 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
1031 processing and selection are described in Supplemental Material. The repos-
1032 itory contains the reproducible analysis and does not redistribute the raw
1033 recordings.

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1037 **Use of AI tools**

1038 AI assistance to this work was substantial and is not separable by task. Sev-
1039 eral coding assistants and chat interfaces were used in an interleaved way
1040 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
1041 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
1042 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
1043 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
1044 accessed with the Hermes CLI. Between them they contributed to the Lean
1045 development, the simulation and analysis code, and the drafting and revision
1046 of both documents. Because the tools were used in combination and in alter-
1047 nation, no section, proof, figure or result is attributable to any one of them,
1048 and none is claimed to be. The author is responsible for all content, including
1049 everything a tool produced. Machine checking establishes the stated Lean
1050 results under their hypotheses; it does not validate biological interpretations
1051 or replace scientific review of generated code and prose.

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